■ Integrated Secure Data Erasure & Advanced File Recovery Tool

SIH 2026 — Digital Forensics & Data Sanitization

An integrated digital forensics prototype designed to securely sanitize data, recover deleted files, verify sa

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■ Problem Statement

Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for

The objective of this project is to build a unified platform that combines:

- Secure data sanitization
- File and folder deletion
- Disk-image sanitization
- Filesystem analysis
- Deleted-file recovery
- Raw file carving
- Fragmented-file reconstruction
- Recovery validation
- Recovery confidence scoring
- Recoverability scoring
- Before/after sanitization comparison
- Sanitization verification
- Tamper-evident audit logging
- Automated forensic reporting
- Sanitization certificate generation

The prototype is designed as a focused **2–3 day MVP** using controlled test disk images.

■ Core Project Objective

The main idea of the project is:

> **Don't just erase data. Prove what was recoverable before sanitization — and prove what remains after

Instead of simply performing data deletion, the system creates a closed forensic loop:

```text
RECOVER
  ↓
MEASURE
  ↓
BASELINE
  ↓
SANITIZE
  ↓
RECOVER AGAIN
  ↓
COMPARE
  ↓
VERIFY
  ↓
REPORT
```

```

This makes the system capable of demonstrating the effectiveness of sanitization using measurable recov

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# ■■ Overall System Architecture

```text
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
               ■     USER / INVESTIGATOR     ■
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                     ■
                     ▼
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
               ■      UNIFIED GUI          ■
               ■  Forensic Data Management   ■
               ■        Platform           ■
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                     ■
           ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
           ■         ■         ■
           ▼         ▼         ▼
   ■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■
   ■    TEAM 1      ■   ■    TEAM 2      ■   ■    TEAM 3      ■
   ■  SANITIZATION  ■   ■  FILE RECOVERY  ■   ■  INTELLIGENCE &  ■
   ■  & FILE ERASE  ■   ■  & FILE CARVING  ■   ■  VERIFICATION  ■
   ■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■
          ■          ■          ■
          ▼          ▼          ▼
    ■■■■■■■■■■■■■■■■    ■■■■■■■■■■■■■■■■    ■■■■■■■■■■■■■■■■■■
    ■ File Eraser  ■    ■ Sleuth Kit  ■    ■ Recovery Score ■
    ■ Folder Eraser■    ■ Recovery    ■    ■ Baseline       ■
    ■ Image Eraser ■    ■ Raw Carving  ■    ■ Comparison     ■
    ■ Media Class. ■    ■ Reconstruction■    ■ Verification   ■
    ■■■■■■■■■■■■■■■■    ■ Validation   ■    ■ Audit Trail    ■
                        ■■■■■■■■■■■■■■■■    ■ Reporting      ■
                                        ■■■■■■■■■■■■■■■■■■
                     ■
                     ▼
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
               ■    CONSOLIDATED RESULT     ■
               ■                         ■
               ■ Recovery + Sanitization +  ■
               ■ Verification + Audit +     ■
               ■ Certificate / Report       ■
               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

■ Team Structure

The project is divided into three major teams:

Team	Module	Primary Responsibility
Team 1	Data Sanitization & File Sanitization	Securely remove data
```

| Team 2 | File Recovery | Find, recover, reconstruct and validate files |
| Team 3 | Algorithm / Forensic Intelligence & Verification | Score, compare, verify, audit and report |

Each team develops its own module while following common:

- Design concepts
- Visual language
- Data structures
- Integration contracts

The project lead connects all modules into one unified prototype.

■ TEAM 1 — DATA SANITIZATION & FILE SANITIZATION

■ Purpose

Team 1 is responsible for securely removing data and implementing the sanitization layer of the prototype.

The sanitization module is capable of handling:

- Individual files
- Folders
- Controlled disk images

It also classifies the target storage medium and recommends an appropriate sanitization approach.

■ Team 1 Modules

1. File Sanitization

Responsible for sanitizing individual files.

Features

- File sanitization/deletion
- Target identification
- SHA-256 hash capture before sanitization
- Sanitization method selection
- Sanitization status
- Timestamp recording
- Explicit confirmation before destructive operation

---

2. Folder Sanitization

Responsible for sanitizing folders and their contents.

Features

- Folder selection
- File enumeration
- File sanitization
- Folder-level result reporting

- Target information
- Timestamp
- Sanitization status

---

3. Controlled Disk-Image Sanitization

Responsible for demonstrating sanitization against controlled forensic disk images.

The MVP should use test disk images rather than automatically sanitizing physical devices.

Example:

```text
Controlled Disk Image
        ■
        ▼
Target Identification
        ■
        ▼
SHA-256 Capture
        ■
        ▼
Sanitization Method
        ■
        ▼
Sanitization
        ■
        ▼
Result
```

---

4. Storage Media Classification

The system identifies the target medium as:

- HDD
- SSD/NVMe
- USB/Flash
- Disk Image
- Unknown

The classifier is used to recommend an appropriate sanitization approach.

---

5. Sanitization Method Recommendation

The system provides a sanitization recommendation based on the target medium.

```text
                Target
                   ■
                   ▼
           Media Classification
```

    ■
■■■■■■■■■■■■■■■■■■■■■
■   ■   ■
▼   ▼   ▼
HDD   SSD   USB/Flash
■   ■   ■
■■■■■■■■■■■■■■■■■■■■■
    ■
    ▼
Recommended Method
```

The MVP primarily demonstrates classification and recommendation for physical media rather than automa

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## 6. Sanitization Safety Layer

Destructive operations must include safety controls.

### Safety requirements

- Explicit user confirmation
- Display target path
- Display SHA-256 before sanitization
- Prevent accidental system-disk targeting
- Controlled test-image operation by default

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# ■ TEAM 2 — FILE RECOVERY

## ■ Purpose

Team 2 is responsible for finding, reconstructing, recovering, and validating deleted or otherwise recoveral

The recovery module combines:

- Sleuth Kit filesystem analysis
- Filesystem metadata analysis
- Deleted-file identification
- Filesystem-based recovery
- Raw file carving
- Fragmented-file reconstruction
- Recovery validation

---

# ■ Team 2 Modules

## 1. Disk Image Analysis

Analyzes the provided forensic disk image.

Responsibilities include:

- Identifying partitions
- Identifying filesystems
```

- Extracting filesystem information
- Locating relevant evidence

---

2. Sleuth Kit Integration

The project uses **The Sleuth Kit (TSK)** for filesystem-level forensic analysis.

Important tools include:

```text
mmls
fsstat
fls
istat
```

General workflow:

```text
Disk Image
    ■
    ▼
   mmls
    ■
    ▼
Partition Information
    ■
    ▼
  fsstat
    ■
    ▼
Filesystem Information
    ■
    ▼
   fls
    ■
    ▼
Deleted File Identification
    ■
    ▼
Metadata / Inode Analysis
    ■
    ▼
Recovery
```

---

3. Deleted File Identification

The system identifies files that are marked as deleted but whose data may still exist in the disk image.

Information may include:

- Filename
- Inode

- File size
- Location
- Allocation status
- Deleted status
- Filesystem evidence

---

4. Filesystem-Based File Recovery

Deleted files identified through filesystem metadata can be recovered using Sleuth Kit-based methods.

```text
Deleted File
   ■
   ▼
Filesystem Metadata
   ■
   ▼
Locate Data Blocks
   ■
   ▼
Extract Data
   ■
   ▼
Recovered Artifact
```

---

5. Raw File Carving

Raw file carving operates directly on the disk-image bytes.

This approach is useful when filesystem metadata is:

- Missing
- Damaged
- Unavailable
- Insufficient

The carving engine searches for file signatures and extracts file data based on file structure.

■ Initial File-Carving Formats

The initial MVP targets:

File Type	Extension
JPEG	`.jpg`, `.jpeg`
PNG	`.png`
PDF	`.pdf`
Microsoft Word	`.docx`
ZIP	`.zip`

■ Raw File Carving Pipeline

```text
Disk Image
  ■
  ▼
Read Raw Bytes
  ■
  ▼
Search File Signatures
  ■
  ▼
Identify File Header
  ■
  ▼
Determine File Structure
  ■
  ▼
Locate Appropriate End Marker
  ■
  ▼
Extract File
  ■
  ▼
Validate File
  ■
  ▼
Generate Metadata
```

---

6. Fragmented File Reconstruction

A file may be stored in multiple non-contiguous locations on a disk.

Example:

```text
[FILE HEADER]
[FILE DATA]
[RANDOM DATA]
[RANDOM DATA]
[FILE DATA]
[FILE DATA]
[FILE END]
```

The system attempts to reconstruct fragmented files where sufficient structural information is available.

Possible evidence:

- File signatures
- File structure
- Filesystem metadata
- Block/cluster information
- Structural markers

- Validation results

---

7. Recovered File Validation

A recovered file is not automatically considered valid just because bytes were extracted.

The validation layer checks whether the recovered data represents a usable file.

Possible checks include:

- File signature verification
- Structural validation
- Parser validation
- Successful decoding
- Successful opening
- File size verification
- SHA-256 calculation
- Integrity status

```text
Recovered Artifact
     ■
     ▼
Signature Check
     ■
     ▼
Structure Check
     ■
     ▼
Parser Validation
     ■
     ▼
Integrity Check
     ■
     ▼
VALID / PARTIAL / INVALID
```

---

8. Recovery Metadata

Each recovered artifact should contain metadata such as:

```json
{
   "file_name": "recovered_001.jpg",
   "file_type": "JPEG",
   "source_image": "carving.img",
   "offset": 1048576,
   "size": 245760,
   "recovery_method": "raw_carving",
   "validation_status": "VALID",
   "sha256": "..."
}
```

This metadata becomes evidence for Team 3.

■ TEAM 3 — ALGORITHM / FORENSIC INTELLIGENCE & VERIFICATION

■ Purpose

Team 3 converts recovery and sanitization results into measurable and explainable forensic evidence.

Team 3 is responsible for:

- Recovery confidence
- Recoverability scoring
- Baseline creation
- Before/after comparison
- Sanitization verification
- Audit trail
- Reporting
- Certificate generation

■ Team 3 Modules

1. Explainable Recovery Confidence

The system produces a recovery confidence score supported by understandable evidence.

Possible evidence includes:

- Valid file signature
- Structural checks
- Parser validation
- Filesystem evidence
- Successful file decoding
- Integrity checks

The goal is not just:

```text
FILE FOUND
```

but:

```text
FILE FOUND
   ■
   ■■■ Valid Signature
   ■■■ Valid Structure
   ■■■ Parser Passed
   ■■■ Filesystem Evidence
   ■■■ Integrity Passed
         ■
         ▼
    RECOVERY CONFIDENCE
```

```

---

# 2. Recoverability Score

The project defines a **Recoverability Score**.

This score summarizes how recoverable an artifact is rather than simply reporting that the file was found.

Example conceptual model:

```text
Filesystem Evidence
        +
Signature Validity
        +
Structural Validity
        +
Parser Validation
        +
Integrity
        ■
        ▼
Recoverability Score
```

The exact scoring algorithm is project-defined.

---

3. Pre-Sanitization Recovery Baseline

Before sanitization, the system performs recovery and records the identified artifacts.

Example:

```text
                  BEFORE SANITIZATION
                          ■
                          ▼
                    Analyze Image
                          ■
                          ▼
                    Recover Files
                          ■
                          ▼
                     Validate Files
                          ■
                          ▼
                   Calculate Scores
                          ■
                          ▼
                  CREATE BASELINE
```

The baseline records what was recoverable before sanitization.
```

---

4. Post-Sanitization Recovery

After Team 1 sanitizes the target, Team 2 performs recovery again.

```text
Baseline
   ■
   ▼
Sanitization
   ■
   ▼
Post-Sanitization Image
   ■
   ▼
Recovery
   ■
   ▼
Validation
   ■
   ▼
Post-Sanitization Results
```

---

5. Before/After Comparison

Team 3 compares:

```text
BEFORE SANITIZATION
        ■
        ▼
Recovered Artifacts
        ■
        ▼
Recovery Baseline
        ■
        ■
        ▼
SANITIZATION
        ■
        ▼
AFTER SANITIZATION
        ■
        ▼
Recovered Artifacts
        ■
        ▼
Comparison
```

The system determines whether previously identified artifacts remain recoverable.

---

6. Sanitization Verification

The system verifies the sanitization outcome based on the defined recovery procedure.

The verification statement is:

> **"No previously identified artifacts were recovered under the defined verification procedure."**

The system can produce:

```text
BEFORE
  ■
  ■■■ Artifact A
  ■■■ Artifact B
  ■■■ Artifact C
      ■
      ▼
  SANITIZATION
      ■
      ▼
AFTER
  ■
  ■■■ Artifact A ■
  ■■■ Artifact B ■
  ■■■ Artifact C ■
      ■
      ▼
    VERIFICATION
      ■
      ▼
      PASS
```

If previously identified artifacts remain recoverable, the verification result can be marked as **FAIL**.

---

7. Tamper-Evident Audit Trail

The system maintains a hash-chained audit trail.

Example:

```text
Event 1
  ■
  ▼
Hash 1
  ■
  ▼
Event 2 + Hash 1
  ■
  ▼
Hash 2
  ■
  ▼
Event 3 + Hash 2
```

■
▼
Hash 3
```

If an earlier recorded event is modified, the hash chain becomes inconsistent.

This provides an integrity mechanism for the recorded workflow.

> This is a tamper-evident audit mechanism and **not blockchain**.

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# 8. Automated Forensic Report

The system generates a consolidated report containing:

- Target information
- Recovery findings
- Recovery metadata
- Recovery confidence
- Recoverability score
- Sanitization details
- Before/after results
- Verification outcome
- Audit integrity
- Final status

---

# 9. Sanitization Certificate

The system can generate a sanitization certificate/report containing the final verification outcome.

Example:

```text
==================================================
        SANITIZATION CERTIFICATE
==================================================

Target            : carving.img

Pre-Sanitization
Artifacts Recovered : 5

Sanitization Status : COMPLETED

Post-Sanitization
Artifacts Recovered : 0

Verification       : PASS

Audit Integrity     : VALID

Statement:
No previously identified artifacts were recovered
under the defined verification procedure.

==================================================
```

■ COMPLETE SYSTEM WORKFLOW

The complete integrated workflow is:

```text
          ■■■■■■■■■■■■■■■■■■■■■■
          ■  TEST DISK IMAGE   ■
          ■■■■■■■■■■■■■■■■■■■■■■
              ■
              ▼
            ■■■■■■■■■■■
            ■ ANALYZE ■
            ■■■■■■■■■■■
              ■
              ▼
          ■■■■■■■■■■■■■■■■■■■■■■
          ■ FIND DELETED FILES ■
          ■■■■■■■■■■■■■■■■■■■■■■
              ■
              ▼
            ■■■■■■■■■■■
            ■ RECOVER ■
            ■■■■■■■■■■■
              ■
              ▼
            ■■■■■■■■■■■■
            ■ VALIDATE ■
            ■■■■■■■■■■■■
              ■
              ▼
        ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
        ■ CALCULATE RECOVERY         ■
        ■ CONFIDENCE & SCORE         ■
        ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
              ■
              ▼
          ■■■■■■■■■■■■■■■■■■■■
          ■ CREATE BASELINE  ■
          ■■■■■■■■■■■■■■■■■■■■
             ■
             ▼
           ■■■■■■■■■■■■■■
           ■ SANITIZE   ■
           ■■■■■■■■■■■■■■
             ■
             ▼
          ■■■■■■■■■■■■■■■■■■■
          ■ RECOVER AGAIN   ■
          ■■■■■■■■■■■■■■■■■■■
             ■
             ▼
         ■■■■■■■■■■■■■■■■■■■■■

```
      ■ COMPARE BEFORE /  ■
      ■ AFTER           ■
      ■■■■■■■■■■■■■■■■■■■■■
          ■
          ▼
        ■■■■■■■■■■■■
        ■ VERIFY   ■
        ■■■■■■■■■■■■
          ■
          ▼
     ■■■■■■■■■■■■■■■■■■■■■■■
     ■ GENERATE REPORT &   ■
     ■ CERTIFICATE        ■
     ■■■■■■■■■■■■■■■■■■■■■■■
```

■ TEAM INTEGRATION

The three teams operate as a closed-loop system.

```text
          TEAM 2
       FILE RECOVERY
          ■
          ■ Recovery Evidence
          ▼
          TEAM 3
    INTELLIGENCE & VERIFY
          ■
          ■ Baseline
          ▼
          TEAM 1
       SANITIZATION
          ■
          ■ Sanitized Target
          ▼
          TEAM 2
       FILE RECOVERY
          ■
          ■ Post-Recovery Evidence
          ▼
          TEAM 3
    COMPARE & VERIFY
          ■
          ▼
        FINAL REPORT
```

■ Integration Responsibility

Stage	Responsible Team	Main Output
Analyze test image	Team 2	Filesystem/evidence information
Find deleted files	Team 2	Recovered artifacts

Validate recovery	Team 2	Validated recovery results
Calculate confidence/score	Team 3	Confidence + recoverability metrics
Create baseline	Team 3	Pre-sanitization evidence baseline
Sanitize target	Team 1	Sanitization result
Recover again	Team 2	Post-sanitization recovery results
Verify & compare	Team 3	Before/after verification
Generate report	Team 3	Tamper-evident report/certificate

■■ UNIFIED GUI

The final prototype is intended to expose all major capabilities through one unified interface.

Suggested navigation:

```text
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■           SECURE FORENSIC PLATFORM           ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                                    ■
■  Dashboard                                 ■
■                                    ■
■  ■■■ Evidence / Disk Image               ■
■  ■                                 ■
■  ■■■ File Recovery                       ■
■  ■     ■■■ Filesystem Analysis             ■
■  ■     ■■■ Deleted Files                 ■
■  ■     ■■■ Raw File Carving               ■
■  ■     ■■■ Reconstruction                 ■
■  ■     ■■■ Validation                   ■
■  ■                                 ■
■  ■■■ Sanitization                       ■
■  ■     ■■■ File Sanitization               ■
■  ■     ■■■ Folder Sanitization             ■
■  ■     ■■■ Disk Image Sanitization           ■
■  ■     ■■■ Media Classification             ■
■  ■                                 ■
■  ■■■ Analysis & Verification               ■
■  ■     ■■■ Recovery Confidence               ■
■  ■     ■■■ Recoverability Score             ■
■  ■     ■■■ Baseline                     ■
■  ■     ■■■ Before/After Comparison           ■
■  ■     ■■■ Sanitization Verification         ■
■  ■                                 ■
■  ■■■ Audit Trail                       ■
■  ■                                 ■
■  ■■■ Reports & Certificates               ■
■                                    ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

■ Suggested Repository Structure

```text
integrated-secure-forensic-tool/
```

```
■
■■■ README.md
■
■■■ frontend/
■   ■■■ dashboard/
■   ■■■ recovery/
■   ■■■ sanitization/
■   ■■■ verification/
■   ■■■ audit/
■   ■■■ reports/
■
■■■ backend/
■   ■■■ api/
■   ■■■ recovery/
■   ■■■ sanitization/
■   ■■■ verification/
■   ■■■ audit/
■   ■■■ reporting/
■
■■■ recovery/
■   ■■■ sleuthkit/
■   ■■■ carving/
■   ■■■ reconstruction/
■   ■■■ validation/
■   ■■■ signatures/
■
■■■ sanitization/
■   ■■■ file/
■   ■■■ folder/
■   ■■■ disk-image/
■   ■■■ media-classifier/
■
■■■ intelligence/
■   ■■■ confidence/
■   ■■■ scoring/
■   ■■■ baseline/
■   ■■■ comparison/
■
■■■ audit/
■   ■■■ hash-chain/
■
■■■ reports/
■
■■■ certificates/
■
■■■ test-data/
■
■■■ disk-images/
■
■■■ docs/
```

> The final repository structure can be modified according to the implementation architecture.

■ TESTING STRATEGY

The MVP should primarily use controlled test disk images.

Test Scenario 1 — Deleted File Recovery

```text
Create File
    ■
    ▼
Store File
    ■
    ▼
Delete File
    ■
    ▼
Create Disk Image
    ■
    ▼
Analyze Image
    ■
    ▼
Find Deleted File
    ■
    ▼
Recover
    ■
    ▼
Validate
```

Test Scenario 2 — Raw File Carving

```text
Create Test File
    ■
    ▼
Place Data in Disk Image
    ■
    ▼
Remove Filesystem Reference
    ■
    ▼
Raw Scan
    ■
    ▼
Detect Signature
    ■
    ▼
Carve
    ■
    ▼
Validate
```

Test Scenario 3 — Sanitization Verification

```text
Create Test Image
       ■
       ▼
Recover Files
       ■
       ▼
Create Baseline
       ■
       ▼
Sanitize Image
       ■
       ▼
Recover Again
       ■
       ▼
Compare Results
       ■
       ▼
Verify
       ■
       ▼
Generate Certificate
```

Test Scenario 4 — Multiple File Formats

The test image can contain:

```text
carving.img
■
■■■ image.jpg
■■■ image.png
■■■ document.pdf
■■■ document.docx
■■■ archive.zip
```

The recovery engine attempts to identify and recover each supported artifact.

■ Example Final Result

```text
============================================================
                FORENSIC ANALYSIS RESULT
============================================================

TARGET
------------------------------------------------------------
Disk Image          : carving.img
SHA-256             : <hash>
```

```
PRE-SANITIZATION
-------------------------------------------------------
Files Identified    : 5
Files Recovered     : 5
Files Validated     : 5

Recovery Confidence : <score>
Recoverability      : <score>

SANITIZATION
-------------------------------------------------------
Method              : <method>
Status              : COMPLETED

POST-SANITIZATION
-------------------------------------------------------
Files Recovered     : 0
Previously Identified
Artifacts Recovered : 0

VERIFICATION
-------------------------------------------------------
Result              : PASS

Statement:
No previously identified artifacts were recovered
under the defined verification procedure.

AUDIT
-------------------------------------------------------
Audit Chain         : VALID

FINAL STATUS
-------------------------------------------------------
SANITIZATION        : VERIFIED
=======================================================
```

■ SECURITY & SAFETY BOUNDARY

The MVP should operate on **controlled test disk images by default**.

Mandatory Safety Rules

- Never automatically target system disks.
- Never automatically target `/dev/sda`.
- Never automatically target `/dev/nvme*`.
- Destructive operations require explicit confirmation.
- Display the target path before sanitization.
- Display the SHA-256 hash before sanitization.
- Use controlled test images for destructive demonstrations.
- Do not perform destructive operations on real user data.

■■ SSD / NVMe Limitation

The prototype should not claim universal or permanent deletion, particularly for:

- SSD
- NVMe
- Flash-based storage

For physical storage types, the MVP should primarily demonstrate:

```text
MEDIA CLASSIFICATION
        +
METHOD RECOMMENDATION
```

rather than automatically wiping real physical devices.

■ AI/ML Scope

AI/ML is intentionally **not included in the working MVP** due to the short development window.

It can be presented as a future enhancement.

Potential future applications include:

- Intelligent artifact classification
- Advanced recovery prioritization
- Fragmented-file prediction
- Automated evidence classification
- Anomaly detection
- Recovery success prediction

■ PROJECT UNIQUE SELLING POINTS

1. Primary USP

> **Don't just erase data. Prove what was recoverable before sanitization — and prove what remains after

---

2. Explainable Recovery Confidence

Recovery confidence is supported by understandable forensic evidence such as:

- Valid file signatures
- Structural checks
- Parser validation
- Filesystem evidence

---

3. Recoverability Score

A project-defined metric that describes how recoverable an artifact is rather than simply reporting that a fil

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4. Before/After Verification

The system establishes a recovery baseline before sanitization and compares it with post-sanitization reco

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5. Tamper-Evident Audit Trail

Hash-chained audit events make changes to recorded workflow information detectable.

This is an audit-integrity mechanism and **not blockchain**.

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6. Automated Reporting & Certificate

The system generates a consolidated result containing:

- Target information
- Sanitization details
- Recovery findings
- Verification outcome
- Audit integrity
- Certificate/report

■ MVP Scope

The working MVP focuses on:

Team 1

- File sanitization
- Folder sanitization
- Controlled disk-image sanitization
- Storage-media classification
- Sanitization method recommendation
- Safety controls
- SHA-256 capture
- Sanitization reporting

Team 2

- Sleuth Kit integration
- Filesystem analysis
- Deleted-file identification
- Deleted-file recovery
- Raw file carving
- JPEG recovery
- PNG recovery
- PDF recovery
- DOCX recovery
- ZIP recovery
- Fragmented-file reconstruction

- Recovery validation
- Recovery metadata

Team 3

- Explainable recovery confidence
- Recoverability Score
- Pre-sanitization baseline
- Post-sanitization comparison
- Sanitization verification
- PASS/FAIL determination
- Hash-chained audit trail
- Audit integrity verification
- Automated forensic report
- Sanitization certificate

■ MVP Limitations

The prototype does not claim:

- Universal file recovery
- Guaranteed fragmented-file recovery
- Universal filesystem support
- Permanent deletion on all storage technologies
- Automatic wiping of real physical system drives
- Commercial forensic-suite capabilities
- AI/ML-based recovery during the MVP

■ Future Enhancements

Possible future improvements include:

- More filesystem support
- More file formats
- Advanced fragmented-file reconstruction
- Advanced file parsers
- Parallel carving
- Large-image optimization
- Duplicate artifact detection
- Advanced evidence visualization
- Advanced recovery scoring
- AI/ML-assisted artifact classification
- Additional storage sanitization standards
- Enterprise-scale reporting
- Cloud-based forensic investigation
- Case management
- Multi-user investigator workflows

■■ Technology Stack

The prototype may use the following technologies:

Operating System

- Linux
- Ubuntu
- WSL2

Digital Forensics

- The Sleuth Kit
- `mmls`
- `fsstat`
- `fls`
- `istat`

File Recovery

- Raw byte scanning
- File signatures
- Magic bytes
- File structure analysis
- File carving
- File reconstruction
- File validation

Security

- SHA-256
- Hash-chained audit logs
- Integrity verification

Application

- Unified GUI
- Backend/API
- JSON-based metadata
- Automated reporting

■ Evidence & Audit Model

The system maintains evidence throughout the entire workflow.

```text
TARGET
  ■
  ■■■ Target Path
  ■■■ Target Type
  ■■■ SHA-256
        ■
        ▼
RECOVERY EVIDENCE
  ■
  ■■■ File Type
  ■■■ Offset
  ■■■ Size
  ■■■ Recovery Method
  ■■■ Validation
```

  ■■■ Hash
     ■
     ▼
SANITIZATION RESULT
  ■
  ■■■ Method
  ■■■ Status
  ■■■ Timestamp
     ■
     ▼
POST-RECOVERY EVIDENCE
     ■
     ▼
COMPARISON
     ■
     ▼
VERIFICATION
     ■
     ▼
AUDIT TRAIL
     ■
     ▼
FINAL REPORT
```

---

# ■ Final System Concept

The entire platform can be summarized as:

```text
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■               INTEGRATED FORENSIC TOOL                ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                                               ■
■                 CONTROLLED TEST IMAGE                 ■
■                        ■                      ■
■                        ▼                      ■
■                   FILE RECOVERY               ■
■                        ■                      ■
■                ■■■■■■■■■■■■■■■■■■■■■                ■
■                ■       ■        ■             
■                ▼       ▼        ■             
■             SLEUTH KIT      RAW CARVING               ■
■                ■       ■        ■             
■                ■■■■■■■■■■■■■■■■■■■■■                ■
■                        ▼                      ■
■                     VALIDATION                ■
■                        ■                      ■
■                        ▼                      ■
■                 RECOVERY CONFIDENCE                   ■
■                        ■                      ■
■                        ▼                      ■
■                     BASELINE                  ■
■                        ■                      ■
■                        ▼                      ■
■                   SANITIZATION                ■

```
■                    ■                    ■
■                    ▼                    ■
■             RECOVERY AGAIN                        ■
■                    ■                    ■
■                    ▼                    ■
■             BEFORE / AFTER                      ■
■               COMPARISON                        ■
■                    ■                    ■
■                    ▼                    ■
■                  VERIFY                   ■
■                    ■                    ■
■                    ▼                    ■
■             AUDIT + REPORT                      ■
■                    ■                    ■
■                    ▼                    ■
■           SANITIZATION CERTIFICATE                ■
■                                        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

■■■ Team Responsibilities Summary

```text
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                  TEAM 1                   ■
■             DATA SANITIZATION                     ■
■                                        ■
■  FILE ERASE → FOLDER ERASE → IMAGE SANITIZATION        ■
■            → MEDIA CLASSIFICATION                ■
■            → METHOD RECOMMENDATION                ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                  ■
                  ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                  TEAM 2                   ■
■               FILE RECOVERY                     ■
■                                        ■
■  ANALYZE → FIND → RECOVER → CARVE → RECONSTRUCT        ■
■               → VALIDATE                  ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                  ■
                  ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                  TEAM 3                   ■
■          FORENSIC INTELLIGENCE & VERIFICATION         ■
■                                        ■
■  SCORE → BASELINE → COMPARE → VERIFY → AUDIT → REPORT     ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

■ Final Outcome

The final prototype brings all three teams together into one closed-loop forensic platform.

```text
RECOVER WHAT EXISTS
        ■
        ▼
MEASURE RECOVERY
        ■
        ▼
RECORD BASELINE
        ■
        ▼
 SANITIZE DATA
        ■
        ▼
ATTEMPT RECOVERY
        ■
        ▼
COMPARE RESULTS
        ■
        ▼
 VERIFY RESULT
        ■
        ▼
PROVE THE OUTCOME
        ■
        ▼
GENERATE REPORT/CERTIFICATE
```

The goal is not merely to erase data or recover files independently.

The goal is to provide an **evidence-driven, measurable, and verifiable workflow for data sanitization and f

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■ License

This project is developed for educational, research, and **Smart India Hackathon (SIH) 2026** prototype p